

Equivalence of inequality indices in the three-dimensional model of informetric impact

Lucio Bertoli-Barsotti^a, Marek Gagolewski^{b,c,*}, Grzegorz Siudem^d and Barbara Żogała-Siudem^c

^aUniversity of Bergamo, Department of Economics, Italy

^bWarsaw University of Technology, Faculty of Mathematics and Information Science, ul. Koszykowa 75, 00-662 Warsaw, Poland

^cSystems Research Institute, Polish Academy of Sciences, ul. Newelska 6, 01-447 Warsaw, Poland

^dWarsaw University of Technology, Faculty of Physics, ul. Koszykowa 75, 00-662 Warsaw, Poland

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Abstract

Inequality is an inherent part of our lives: we see it in the distribution of incomes, talents, citations, to name a few. However, its intensity varies across environments: there are systems where the available resources are relatively evenly distributed but also where a small group of items or agents controls the majority of assets. Numerous indices for quantifying the degree of inequality have been proposed but in general, they work quite differently.

We recently observed (Siudem et al., PNAS 117:13896–13900, 2020) that many rank-size distributions might be approximated by a time-dependent agent-based model involving a mixture of preferential (rich-get-richer) and accidental (sheer chance) attachment. In this paper, we point out its relationship to an iterative process that generates rank distributions of any length and a predefined level of inequality, as measured by the Gini index.

We prove that, under our model, the Gini, Bonferroni, De Vergottini, and Hoover indices are equivalent for samples of similar sizes. Given one of them, we can recreate the value of another measure. Thanks to the obtained formulae, we can also understand how they depend on the sample size. An empirical analysis of a large database of citation records in economics (RePEc) yields a good match with our theoretical derivations.

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1. Introduction

Given a series of measurements, indicators, scores, counts, or any other numeric values, it is natural to order them from the highest to the lowest. This way, we get better insight into the aspects of reality they aim to capture. In particular, various rankings (e.g., of universities, movies, or restaurants; see Iñiguez et al., 2022) promise to make our lives easier by claiming they can separate seeds from the chaff. Investigation and prediction of the size of the top or otherwise extreme values is crucial in risk analysis or disaster prevention (Voitalov et al., 2019; Pickands III, 1975; Marshall and Olkin, 2007). The search for patterns and universalities in sorted data remains a fundamental, multidisciplinary research topic (Holme, 2022; Newman, 2005). This includes the study of ranking dynamics (Iñiguez et al., 2022) and the distribution of their static snapshots, from the most straightforward Zipf power-law models (Newman, 2005) to more complex ones (Petersen et al., 2011; Siudem et al., 2020, 2022; Singh et al., 2022). Rankings are inextricably linked to inequality. Unfortunately, not everyone can stand on the podium; often, the winner takes it all (Perc, 2014), especially in the context of skewed distributions commonly observed in informetrics (Price, 1965; Newman, 2005). To quote Rousseau et al. (2018, p. 308) “the informetric laws describe situations in which a large inequality is present”. For example, long-tailed distributions, such as those described by the power laws (Newman, 2005) or Pareto-type distributions (Arnold, 2015; Siudem et al., 2022), are typically used in informetrics to model information production processes (IPPs; systems consisting of “sources” producing “items”; Egghe and Rousseau, 1990; Egghe, 2005) where

*Corresponding author

Email addresses: lucio.bertoli-barsotti@unibg.it (L. Bertoli-Barsotti); marek.gagolewski@pw.edu.pl (M. Gagolewski); grzegorz.siudem@pw.edu.pl (G. Siudem); zogala@ibspan.waw.pl (B. Żogała-Siudem)

URL: <https://www.gagolewski.com> (M. Gagolewski); <http://if.pw.edu.pl/~siudem> (G. Siudem)

ORCID(s): 0000-0003-4508-3172 (L. Bertoli-Barsotti); 0000-0003-0637-6028 (M. Gagolewski); 0000-0002-9391-6477 (G. Siudem); 0000-0002-2869-7300 (B. Żogała-Siudem)

we observe a high concentration of assets amongst only a few top scorers. A concentration measure is one of the two elements, the other being *the production* of the theory on impact (Egghe and Rousseau, 2022, 2023a), and the Lorenz order is the most direct way to represent the concentration relationship – which is ultimately a method for establishing rankings among IPPs.

The recently-proposed 3DI model (three dimensions of impact; Siudem et al., 2020, 2022), can be conceived of as a rank-size approach to describing the mechanisms governing the growth of bibliographic and other networks studied originally by Price (1965). Namely, consider a process where, in every time step, a system (e.g., a citation network, a cluster of internet portals) grows by one entity (e.g., a new paper, a website). In each iteration, we distribute m impact or wealth units (e.g., citations, links) amongst the already-existing entities:

- $a = (1 - \rho)m$ units totally at random,
- $p = \rho m$ units according to the preferential attachment rule,

with ρ representing the extent to which the rich-get-richer rule dominates over pure luck. Inspired by the observation in (Bertoli-Barsotti, 2023; Bertoli-Barsotti et al., 2024), in Section 2, we will show that the ρ parameter naturally corresponds to the value of the Gini index of the resulting ordered sample of impact measures. We will thus indicate the relationship between the degree of preferentiality and inequality.

Different systems or environments naturally have varied sizes and levels of inequality (Cowell, 2000; Silber, 2012), i.e., what percentage of the top scorers is in possession of the majority of the resources. As the degree of evenness vs monopoly can be measured by different indicators (e.g., the Gini, Bonferroni, or Hoover index), a question of which one is the most informative often arises (Prathap, 2014, 2022). In Section 3, we will show the equivalence of many popular inequality indices in our model by expressing them in terms of monotone, 1-to-1 functions of the Gini index.

In Section 4, an analysis of a large number of citation records consisting of research papers in economics (RePEc) will show a quite good match between the empirical data and our theoretical derivations.

2. The Gini-stable process, rich-get-richer, and random distribution of wealth

In this section, after introducing some basic notation convention, we recall the iterative affine process proposed by Bertoli-Barsotti (2023), which Bertoli-Barsotti et al. (2024) proved to preserve the Gini index throughout all its iterations. Moreover, it is the only such process, as shown in Section 2.4. In Section 2.5, we prove that it is equivalent to the 3DI model by Siudem et al. (2020) but in a different parametrisation, given by the Gini index. In Section 2.6, we connect the normalised vectors $\mathbf{p}^{(N,G)}$ with the Lorenz curves.

2.1. Notation

We utilise the following notation. The gamma function is given by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$, $\Gamma(z+1) = z\Gamma(z)$, the polygamma functions are defined by $\psi^{(m)}(z) = \frac{d^{m+1}}{dz^{m+1}} \log(\Gamma(z))$, the digamma function is $\psi(z) = \psi^{(0)}(z) = \Gamma'(z)/\Gamma(z)$, the harmonic number $H_n = \sum_{k=1}^n k^{-1} = \psi(n+1) - \gamma$, where $\gamma \approx 0.577$ is the Euler constant.

2.2. The derivation of the Gini-stable process

For any fixed parameter $G \in [0, 1)$, consider an iterative process discussed in (Bertoli-Barsotti et al., 2024) whose update formula features a simple affine function of the consecutive elements:

$$p_k^{(N,G)} = a_N(G) + b_N(G) p_k^{(N-1,G)}, \quad k = 1, \dots, N, \quad (1)$$

for $N \geq 2$, under the assumption that $p_N^{(N-1,G)} = 0$ and $p_1^{(1,G)} = 1$ and:

$$a_N(G) = \frac{1-G}{G(N-2)+1} \frac{1}{N}, \quad (2)$$

$$b_N(G) = \frac{G(N-1)}{G(N-2)+1}. \quad (3)$$

Then, for any $N \geq 2$, $\mathbf{p}^{(N,G)} = (p_1^{(N,G)}, \dots, p_N^{(N,G)})$ is an *ordered normalised positive N -vector*, i.e., $\sum_{k=1}^N p_k^{(N,G)} = 1$ and $1 \geq p_1^{(N,G)} \geq p_2^{(N,G)} \geq \dots \geq p_N^{(N,G)} \geq 0$.

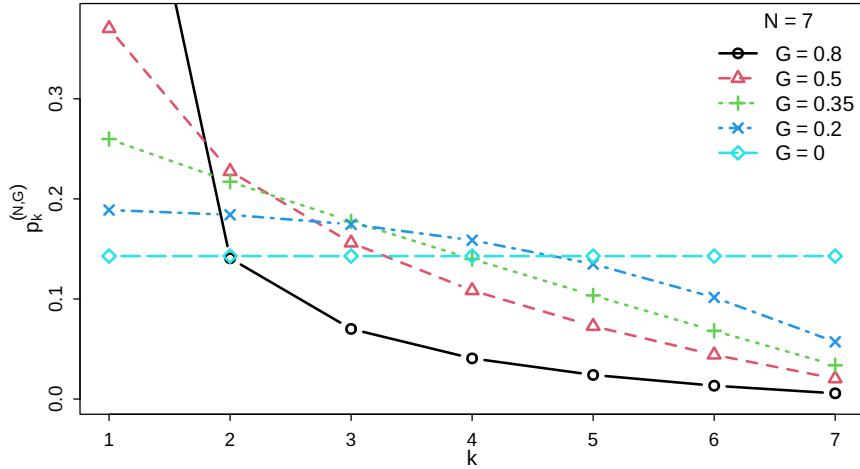


Figure 1: Example ordered normalised N -vectors generated using the affine update formula given by Eq. (1). Parameter G controls the level of inequality, as measured by the Gini index.

Example 1. Here are two example outcomes for $G = 0.25$ and $G = 0.75$ up to $N = 4$:

	$\mathbf{p}^{(N,0.25)}$	$\mathbf{p}^{(N,0.75)}$
$N = 2$	(0.625, 0.375)	(0.875, 0.125)
$N = 3$	(0.450, 0.350, 0.200)	(0.798, 0.155, 0.048)
$N = 4$	(0.350, 0.300, 0.225, 0.125)	(0.743, 0.164, 0.068, 0.025)

There exists an explicit formula for the individual components of the normalised vectors. Namely, we can show that (see Sec. 2.5 and Bertoli-Barsotti et al., 2024):

$$p_k^{(N,G)} = \begin{cases} \frac{1}{N} \frac{1-G}{2G-1} \left(\frac{\Gamma(N+1)}{\Gamma(N+1+1/G-2)} \frac{\Gamma(k+1/G-2)}{\Gamma(k)} - 1 \right), & \text{if } G \neq \frac{1}{2}, \\ \frac{1}{N} (H_N - H_{k-1}) = \frac{1}{N} \sum_{i=k}^N \frac{1}{i}, & \text{if } G = \frac{1}{2}. \end{cases} \quad (4)$$

2.3. Gini index is exactly G .

Figure 1 depicts a few example normalised vectors $\mathbf{p}^{(7,G)}$ for different G s. We see that the G parameter controls the level of inequality of the data distribution: $G = 0$ yields all elements equal to $1/N$, and, as $G \rightarrow 1$, all mass is transferred to the first element.

It turns out that our process generates ordered normalised vectors whose Gini's index, $\mathcal{G}(\mathbf{p}^{(N,G)})$ (Gini, 1912; see also, e.g., Yitzhaki and Schechtman, 2013), is exactly equal to G . Namely, for $N \geq 2$ and $G \neq \frac{1}{2}$ we have:

$$\begin{aligned} \mathcal{G}(p_1^{(N,G)}, \dots, p_N^{(N,G)}) &= \frac{1}{N-1} \sum_{k=1}^N (N-2k+1) p_k^{(N,G)} \\ &= \frac{1-G}{(N-1)(2G-1)} \frac{G(2G-1)(N-1)}{1-G} = G, \end{aligned}$$

and for $G = \frac{1}{2}$ it holds:

$$\mathcal{G}(p_1^{(N,1/2)}, \dots, p_N^{(N,1/2)}) = \frac{(N-1)N}{2(N-1)N} = \frac{1}{2}.$$

Thus, quite remarkably, N (size) and G (inequality) are two *independent* parameters in our model. This is why we refer to the above as the *Gini-stable process*.

2.4. It is the only affine process of this kind.

Our process transforms an ordered normalised vector to a new, longer, ordered normalised vector of the same Gini index. What is more, the aforementioned coefficients a_N and b_N make up the only affine transformation that achieves this property.

Proposition 1. *For any $N \geq 3$ and $G \in [0, 1)$, let $\mathbf{q}^{(N-1)}$ be an ordered probability $(N-1)$ -vector with the Gini index of $\mathcal{G}(q_1^{(N-1)}, \dots, q_{N-1}^{(N-1)}) = G$. Then, generating the normalised N -vector $\mathbf{q}^{(N)}$ using the update formula given by Eq. (1) yields $\mathcal{G}(q_1^{(N)}, \dots, q_N^{(N)}) = G$ if and only if a_N and b_N are given exactly by Eq. (2) and Eq. (3), respectively.*

Proof. We have $\mathcal{G}(q_1^{(N-1)}, \dots, q_{N-1}^{(N-1)}) = \frac{1}{N-2} \sum_{k=1}^{N-1} (N-2k) q_k^{(N-1)} = G$. Moreover,

$$\begin{aligned} \mathcal{G}(q_1^{(N)}, \dots, q_N^{(N)}) &= \frac{1}{N-1} \sum_{k=1}^N (N-2k+1) q_k^{(N)} \\ &= \frac{1}{N-1} \sum_{k=1}^N (N-2k+1) (a_N + b_N q_k^{(N-1)}) \\ &= \frac{a_N}{N-1} \sum_{k=1}^N (N-2k+1) + \frac{b_N}{N-1} \sum_{k=1}^{N-1} (N-2k+1) q_k^{(N-1)} \\ &= 0 + \frac{b_N}{N-1} \sum_{k=1}^{N-1} (N-2k) q_k^{(N-1)} + \frac{b_N}{N-1} \\ &= b_N \left(\frac{N-2}{N-1} G + \frac{1}{N-1} \right) \end{aligned}$$

is equal to G if and only if $b_N = \frac{G(N-1)}{G(N-2)+1}$, which is exactly Eq. (3).

We also need to have $\sum_{k=1}^N p_k^{(N)} = 1$. But after some simple transformations, one can show that this holds if and only if $a_N = (1 - b_N)/N$, which corresponds to Eq. (2). $\square$

2.5. Relation to the 3DI model.

To strengthen the model's underpinnings, let us return to the 3DI (three dimensions of impact) model (Siudem et al., 2020) mentioned in the introduction. Let $X_k(t)$ denote the impact of the k -th richest entity at time step t (e.g., the number of citations to the k -th most cited paper). We assume $X_k(k-1) = 0$ for every k , i.e., the k -th object enters the system with no impact units. The update formula in our model is a mixture of the accidental and rich-get-richer components governed by the ρ parameter¹:

$$X_k(t) = \underbrace{X_k(t-1)}_{\text{previous value}} + \underbrace{\frac{a}{t}}_{\text{accidental income}} + \underbrace{p \frac{X_k(t-1) + \frac{a}{t}}{(t-1)m + a}}_{\text{preferential gain or loss}}, \quad (5)$$

where $a = (1 - \rho)m$ and $p = \rho m$. Let us note that there is some similarity between the preference/randomness structure in the 3DI model and the SJR indicator (González-Pereira et al., 2010) which is a modification of the PageRank algorithm by Brin and Page (1998).

If we assume $\rho = 0$, then we are only left with the accidental component, and our model reduces to the harmonic one (compare Cena et al., 2022):

¹Siudem et al. (2020, 2022) only studied the case of $\rho \in (0, 1)$. However, let us note that $\rho < 0$ is not only possible but also has a nice interpretation (Gagolewski et al., 2022; Bertoli-Barsotti, 2023). In such a scenario, we initially distribute more than the assumed m citations at random, but then we take away from those who are already rich (rich get less).

$$X_k(t) = m \sum_{i=k}^t \frac{1}{i} = m (H_t - H_{k-1}).$$

Note that “purely accidental” does not mean that every entity ends up with the same amount of wealth, as older agents have had more opportunities to become impactful (“the old get richer”). On the other hand, for $\rho < 1$ and $\rho \neq 0$, the solution is:

$$\begin{aligned} X_k(t) &= m \frac{1-\rho}{\rho} \left(\frac{\Gamma(t+1)}{\Gamma(t+1-\rho)} \frac{\Gamma(k-\rho)}{\Gamma(k)} - 1 \right) \\ &= m \frac{1 - \frac{1}{2-\rho}}{\frac{2}{2-\rho} - 1} \left(\frac{\Gamma(t+1)}{\Gamma(t-1+(2-\rho))} \frac{\Gamma(k-2+(2-\rho))}{\Gamma(k)} - 1 \right). \end{aligned} \quad (6)$$

We, therefore, note that:

$$X_k(t) = m t p_k^{(t,G)} \quad (7)$$

with:

$$\rho(G) = 2 - \frac{1}{G} = \frac{2G-1}{G},$$

or, equivalently:

$$G(\rho) = \frac{1}{2-\rho},$$

and $p_k^{(t,G)}$ given by Eq. (1).

We have thus established an aesthetically pleasing connection between the 3DI model (Siudem et al., 2020) and the Gini-stable process (Bertoli-Barsotti et al., 2024), and hence the degree of randomness in the impact distribution and the Gini index. Let us also note that Bertoli-Barsotti et al. (2024) have also studied the asymptotic behaviour of the Lorenz curves generated by this model, and have shown its relationship to the Pareto Type II, exponential, and scaled beta distributions (Arnold, 2015; Pickands III, 1975; Marshall and Olkin, 2007).

2.6. Lorenz curves

For a given nonincreasing N -vector $\mathbf{y} = (y_1, \dots, y_N)$ with $\sum_{i=1}^N y_i = C$, let $I_{\mathbf{y}}(u)$, $0 \leq u \leq 1$, be the continuous curve connecting the origin $(0,0)$ with the point $(1, C)$ and obtained by joining the points $\left(i/N, \sum_{j=1}^i y_{N-j+1}\right)$ by line segments, where $i = 1, 2, \dots, N$. The function $I_{\mathbf{y}}(u)$ may be viewed as an instance of a non-normalised Lorenz curve (Egghe and Rousseau, 2022, 2023a,b), also referred to a generalised Lorenz curve (Rousseau, 2011). In our model, we know the explicit expression for the function $I_{\mathbf{p}^{(N,G)}}(u)$ because the cumulative sums of elements in $\mathbf{p}^{(N,G)}$ can be written in closed-form. Generalised Lorenz curves can then be used to define a class of global impact measures, understood as functions that preserve the non-normalised dominance order $<$ as defined by Egghe and Rousseau (2023a,b) in such a way that $\mathbf{p}^{(N,G)} < \mathbf{p}^{(N',G')}$ holds if and only if $I_{\mathbf{p}^{(N,G)}}(u) \leq I_{\mathbf{p}^{(N',G')}}(u)$ for every $u \in (0, 1)$. A function γ is called a global impact measure if, for any pair of N -tuples $\mathbf{y}_i = C p_i^{(N,G)}$ and $\mathbf{y}'_i = C' p_i^{(N',G')}$, it satisfies the implication:

$$I_{\mathbf{y}}(u) \leq I_{\mathbf{y}'}(u) \Rightarrow \gamma(\mathbf{y}) \leq \gamma(\mathbf{y}').$$

Note that if $G = G'$ and $C \neq C'$, then the vectors $\mathbf{y}$ and $\mathbf{y}'$ have the same Lorenz curves, but different non-normalised Lorenz curves. As well known, the Lorenz ordering is invariant to scale transformations (Egghe and Rousseau, 1991; Rousseau, 1992). Since in the present study we limit our attention to concentration (and not global impact) measures, it is sufficient to only consider the Lorenz curve in its normalised version. Lorenz curves are the basis of a concentration theory and related acceptable measures of concentration (Rousseau et al., 2018, p. 310; Egghe and Rousseau, 2001).

3. Measuring Inequality

Majorisation order. Given two N -vectors $\mathbf{p}$ and $\mathbf{p}'$ with $\sum_{i=1}^N p_i = \sum_{i=1}^N p'_i$, we define the *majorisation order* $\leq$ in such a way that $\mathbf{p} \leq \mathbf{p}'$ if and only if for all $k = 1, \dots, N$ it holds $F_k \leq F'_k$, where $F_k = \sum_{i=1}^k p_{[i]}$ and $F'_k = \sum_{i=1}^k p'_{[i]}$ are the sums of the k greatest elements in $\mathbf{p}$ and $\mathbf{p}'$, respectively (Marshall et al., 2011). In other words, it is the extension of the standard componentwise relation, $\leq$, applied over the consecutive cumulative sums of ordered items. Let us note that in our model, such cumulative sums can be expressed using the following simple formula:

$$F_k^{(N,G)} = \sum_{i=1}^k p_i^{(N,G)} = \begin{cases} \frac{1-G}{2G-1} \left(\frac{G}{1-G} \frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(k-1+1/G)}{\Gamma(k)} - \frac{k}{N} \right), & \text{if } G \neq \frac{1}{2}, \\ \frac{k}{N} (1 + H_N - H_k), & \text{if } G = \frac{1}{2}. \end{cases} \quad (8)$$

Monotonicity w.r.t. the majorisation order. We can show that for any N , $\mathbf{p}^{(N,G)}$ as a function of G is monotone with respect to the majorisation order.

Theorem 1. For any $G \leq G'$ and N , it holds $\mathbf{p}^{(N,G)} \leq \mathbf{p}^{(N,G')}$.

The proof is given by Bertoli-Barsotti et al. (2024), where they discuss the Gini-stable process in the context of the more general Lorenz ordering.

Inequality indices. Let us now consider any function ϕ that maps the ordered normalised N -vectors to the set of real numbers. We say that ϕ is *Schur-convex*, if for any $\mathbf{p} \leq \mathbf{p}'$ it holds $\phi(\mathbf{p}) \leq \phi(\mathbf{p}')$. This is equivalent to ϕ 's being increasing in each $F_1, \dots, F_N$ when re-expressed in terms of cumulative sums of ordered elements; see (Beliakov et al., 2016).

Any Schur-convex function normalised such that $\phi(1, 0, \dots, 0) = 1$ and $\phi(1/N, \dots, 1/N) = 0$ is called an *inequality index*; see (Shorrocks and Foster, 1987) and (Marshall et al., 2011; Lambert, 2001; Chantreuil and Trannoy, 2011; Zheng, 2007; Bosmans, 2016). The Gini index is an example function fulfilling these properties. Below we recall some other noteworthy inequality indices (e.g., Ciommi et al., 2022; Mehran, 1976; Imedio-Olmedo et al., 2012; McVinish and Lester, 2020; Beliaikov et al., 2016). Then, we derive the formulae for different inequality indices as functions of N and G in our model. This will enable us to compute one index based on any other one.

Bonferroni's index. For any normalised N -vector, the Bonferroni index (Bonferroni, 1930) is given by:

$$\begin{aligned} B(p_1, \dots, p_N) &= \frac{N \sum_{i=1}^N \left(1 - \sum_{j=1}^i \frac{1}{N-j+1} \right) p_i}{N-1} = \frac{N \sum_{i=1}^{N-1} \frac{1}{N-i} \sum_{j=1}^i p_j}{N-1} - N + \frac{N}{N-1} \sum_{i=1}^N \left(1 - \frac{1}{N-i+1} \right) \\ &= \frac{N \sum_{i=1}^N \frac{1}{i} \sum_{j=1}^{N-i} p_j}{N-1} + \frac{N \left(1 - \sum_{j=1}^N \frac{1}{j} \right)}{N-1} = \frac{N}{N-1} \left(\sum_{k=1}^{N-1} \frac{F_k}{N-k} + 1 - H_N \right). \end{aligned}$$

Substituting $F_k = F_k^{(N,G)}$ from Eq. (8) for $\frac{1}{2} \neq G \in (0, 1)$, we get:

$$\begin{aligned} B(\mathbf{p}^{(N,G)}) &= \frac{N}{(N-1)(2G-1)} \left(G \sum_{k=1}^{N-1} \frac{\Gamma(N)\Gamma(k-1+1/G)}{(N-k)\Gamma(N-1+1/G)\Gamma(k)} - \frac{1-G}{N} \sum_{k=1}^{N-1} \frac{k}{N-k} + 1 - H_N \right) \\ &= \frac{N}{N-1} \left(\frac{G}{2G-1} (H_{N+1/G-2} - H_{1/G-1}) - \frac{1-G}{(2G-1)N} (N H_{N-1} - N + 1) + 1 - H_N \right) \\ &= \frac{N}{N-1} \left(\frac{G}{2G-1} (H_{N+1/G-2} - H_{1/G-1}) - \frac{1-G}{2G-1} (H_N - 1) + 1 - H_N \right) \\ &= \frac{N}{N-1} \frac{G}{2G-1} (H_{N+1/G-2} - H_{1/G-1} - H_N + 1) \end{aligned}$$

$$= \frac{N}{N-1} \sum_{k=2}^N \frac{1}{k(k+1/G-2)}.$$

For $G = \frac{1}{2}$, we get:

$$\mathcal{B}(\mathbf{p}^{(N,1/2)}) = \frac{N}{N-1} \sum_{k=2}^N \frac{1}{k^2}.$$

Remark 1. Unlike in the Gini index's case, the Bonferroni index depends on the vector length N . However, in the limit as $N \rightarrow \infty$, we have:

$$\mathcal{B}(\mathbf{p}^{(N,G)}) \xrightarrow{N \rightarrow \infty} \begin{cases} \frac{G}{2G-1} (1 - H_{1/G-1}), & \text{for } G \neq \frac{1}{2}, \\ \frac{\pi^2 - 6}{6}, & \text{for } G = \frac{1}{2}. \end{cases}$$

De Vergottini's index. The De Vergottini index (Vergottini, 1940, 1950) is defined as:

$$\begin{aligned} \mathcal{V}(p_1, \dots, p_N) &= \frac{1}{\sum_{i=2}^N \frac{1}{i}} \left(\sum_{k=1}^N \frac{F_k}{k} - 1 \right) = \frac{1}{\sum_{i=2}^N \frac{1}{i}} \left(\sum_{i=1}^N \sum_{j=i}^N \frac{p_i}{j} - 1 \right) \\ &= \frac{1}{H_N - 1} \sum_{j=1}^N p_j (H_N - H_{j-1} - 1) = 1 - \frac{\sum_{k=1}^N p_k H_{k-1}}{H_N - 1}. \end{aligned}$$

Computing the value of the following sum in our model (taking $p_k = p_k^{(N,G)}$ from Eq. (4) with $\frac{1}{2} \neq G \in (0, 1)$):

$$\sum_{k=1}^N p_k^{(N,G)} H_{k-1} = H_N - 1 + \frac{G \left(\frac{G-1}{N} + 1 - 2G \right)}{(G-1)(1-2G)} - \frac{\Gamma(1/G-2)\Gamma(N)}{\Gamma(N-1+1/G)},$$

leads to:

$$\mathcal{V}(\mathbf{p}^{(N,G)}) = \frac{1}{H_N - 1} \left(\frac{\Gamma(1/G-2)\Gamma(N)}{\Gamma(N-1+1/G)} + \frac{G}{(2G-1)N} - \frac{G}{G-1} \right).$$

Furthermore, for $G = \frac{1}{2}$, we get:

$$\mathcal{V}(\mathbf{p}^{(N,1/2)}) = \frac{N - H_N}{N(H_N - 1)}.$$

Remark 2. For $G < 1$, the De Vergottini index in our model converges to zero as $N \rightarrow \infty$. However, the convergence rate is extremely slow; compare Figure 2.

Hoover's index. The Hoover index (Hoover, 1941; also known as the Robin Hood index) can be thought of as the normalised Manhattan distance to the perfectly equal vector:

$$\mathcal{H}(p_1, \dots, p_N) = \frac{N}{2(N-1)} \sum_{k=1}^N \left| p_k - \frac{1}{N} \right|.$$

We can simplify it by determining:

$$\nu = \max \left\{ j : p_j \geq \frac{1}{N} \right\}, \quad (9)$$

and then writing:

$$\begin{aligned} \mathcal{H}(p_1, \dots, p_N) &= \frac{N}{2(N-1)} \left(\sum_{k=1}^{\nu} \left(p_k - \frac{1}{N} \right) - \sum_{k=\nu+1}^N \left(p_k - \frac{1}{N} \right) \right) \\ &= \frac{N}{2(N-1)} \left(\sum_{k=1}^{\nu} \left(p_k - \frac{1}{N} \right) - \underbrace{\left(\sum_{k=1}^N \left(p_k - \frac{1}{N} \right) - \sum_{k=1}^{\nu} \left(p_k - \frac{1}{N} \right) \right)}_{=0} \right) = \frac{1}{N-1} (N F_{\nu} - \nu). \end{aligned}$$

Moreover, for $\frac{1}{2} \neq G \in (0, 1)$, the above results in:

$$\mathcal{H}(p^{(N,G)}) = \frac{G}{2G-1} \left(\frac{N}{N-1} \frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(\nu-1+1/G)}{\Gamma(\nu)} - \frac{\nu}{N-1} \right).$$

Please note that ν is uniquely determined by the values of G and N . Thus, the above does not introduce new parameters to the model.

Remark 3. For large N and $G \neq \frac{1}{2}$, we can express the solution of the continuous approximation to Eq. (9), i.e., $p_{\nu}^{(N,G)} = 1/N$, as:

$$\nu \approx N \left(\frac{1-G}{G} \right)^{G/(2G-1)};$$

see Siudem et al. (2022) for a proof using a slightly different notation in the case of $G > \frac{1}{2}$, which can be easily extended to $G < \frac{1}{2}$. Thanks to this result, we obtain a compact asymptotic formula for the Hoover index:

$$\mathcal{H}(p^{(N,G)}) \xrightarrow{N \rightarrow \infty} \frac{G}{2G-1} \left(\left(\frac{1-G}{G} \right)^{(1-G)/(2G-1)} - \left(\frac{1-G}{G} \right)^{G/(2G-1)} \right) = \left(\frac{1-G}{G} \right)^{(1-G)/(2G-1)}.$$

Also note that for $G = \frac{1}{2}$, we can obtain ν using the Euler–Maclaren formula, which yields:

$$p_k^{(N,1/2)} \approx \frac{1}{N} \left(\frac{1}{2k} + \frac{1}{2N} - \log(k/N) \right).$$

This allows us to solve equation $p^{(\nu,1/2)} = 1/N$ through the principal branch of the Lambert $\mathcal{W}$ function, denoted $\mathcal{W}_0$, leading to:

$$\nu \approx \frac{1}{2} \left(\mathcal{W}_0 \left(\frac{\exp \left(1 - \frac{1}{2N} \right)}{2N} \right) \right)^{-1} = \frac{1+e}{2e} + \frac{N}{e} + \frac{1-e^2}{8eN} + O(N^{-2}) = \frac{(2N+1)e^{-1}+1}{2} + O(N^{-1}),$$

leading us to:

$$\mathcal{H}(p^{(N,1/2)}) \stackrel{N \gg 1}{\approx} \frac{N}{N-1} F_{\left\lfloor \frac{2Ne^{-1}+e^{-1}+1}{2} \right\rfloor}^{(N,1/2)} - \frac{1}{N-1} \left\lfloor \frac{2Ne^{-1}+e^{-1}+1}{2} \right\rfloor \xrightarrow{N \rightarrow \infty} e^{-1}.$$

The P_q indices. For a fixed $q \in (0, 1)$, the P_q index is defined as:

$$P_q(p_1, \dots, p_N) = \frac{\sum_{j=1}^{\lfloor qN \rfloor} p_j - q}{1 - q} = \frac{F_{\lfloor qN \rfloor} - q}{1 - q}. \quad (10)$$

This index is a normalised version of the percentage of accumulated mass in $q100\%$ of the top elements (compare the famous Pareto 80/20 rule). Note that q is as parameter of the index, and not a new parameter of the model; in this sense, P_q is a whole family of indices.

In our model, for $\frac{1}{2} \neq G \in (0, 1)$, the P_q index is equal to:

$$\begin{aligned} P_q(p^{(N,G)}) &= \frac{G}{(2G-1)(1-q)} \left(\frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(\lfloor qN \rfloor - 1 + 1/G)}{\Gamma(\lfloor qN \rfloor)} - \frac{(1-G)\lfloor qN \rfloor}{GN} \right) - \frac{q}{1-q} \approx \\ &\approx \frac{G}{(2G-1)(1-q)} \left(\frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(\lfloor qN \rfloor - 1 + 1/G)}{\Gamma(\lfloor qN \rfloor)} - q \right). \end{aligned}$$

Furthermore, if $G = \frac{1}{2}$, then it holds:

$$P_q(p^{(N,1/2)}) = \frac{q}{1-q} (H_N - H_{\lfloor qN \rfloor}). \quad (11)$$

Remark 4. In the limit as $N \rightarrow \infty$, we have:

$$P_q(p^{(N,G)}) \xrightarrow{N \rightarrow \infty} \begin{cases} \frac{G}{(2G-1)(1-q)} (q^{1/G-1} - q), & \text{for } G \neq \frac{1}{2}, \\ \frac{-q \log(q)}{1-q}, & \text{for } G = \frac{1}{2}. \end{cases}$$

Indices as functions of one another. To sum up, we have shown that in our model, the formulae for the Bonferroni, De Vergottini, Hoover, and P_q indices can be expressed as the following functions of N and G :

$$\tilde{B}(N, G) = \frac{N}{N-1} \sum_{k=2}^N \frac{1}{k(k+1/G-2)}, \quad (12)$$

$$\tilde{V}(N, G) = \frac{1}{H_N - 1} \left(\frac{\Gamma(1/G-2)\Gamma(N)}{\Gamma(N-1+1/G)} + \frac{G}{(2G-1)N} - \frac{G}{G-1} \right), \quad (13)$$

$$\tilde{H}(N, G) = \frac{N}{N-1} \left(\frac{G}{2G-1} \frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(\nu-1+1/G)}{\Gamma(\nu)} - \frac{\nu}{N} \right), \quad (\text{with } \nu = \max\{j : p_j \geq 1/N\}) \quad (14)$$

$$\tilde{P}_q(N, G) = \frac{G}{(2G-1)(1-q)} \left(\frac{\Gamma(N)}{\Gamma(N-1+1/G)} \frac{\Gamma(qN-1+1/G)}{\Gamma(qN)} - q \right), \quad (15)$$

respectively (for readability, we only included the case of $G \neq 1/2$). Figure 2 depicts them for different N s.

We see that G has a higher sensitivity to inequality for low levels of the inequality scale as compared to the H , V and P_q for q close to 0, and a smaller sensitivity at the opposite end of the inequality scale. This confirms and extends the assessments of Ciommi et al. (2022). Also, B and P_q for q close to 1 evidence a higher sensitivity to inequality as compared to G for low levels of the inequality scale, and quite the opposite for high levels of inequality.

We note that the foregoing are all strictly increasing, continuous functions of G . Thus, based on the derived formulae, all the indices can be expressed as one-to-one functions of one another. For instance, given some value of the Bonferroni index B , we can obtain the underlying $G^* = \tilde{B}^{-1}(N, B)$ and then compute, say, $\tilde{V}(N, G^*)$. Even though the analytic formulae for the inverses do not exist, they can easily be solved numerically. In this sense, we can say that – in our model – all the aforementioned inequality indices are equivalent. Similar derivations can be performed for many other inequality indices, although they might not necessarily enjoy analytic solutions.

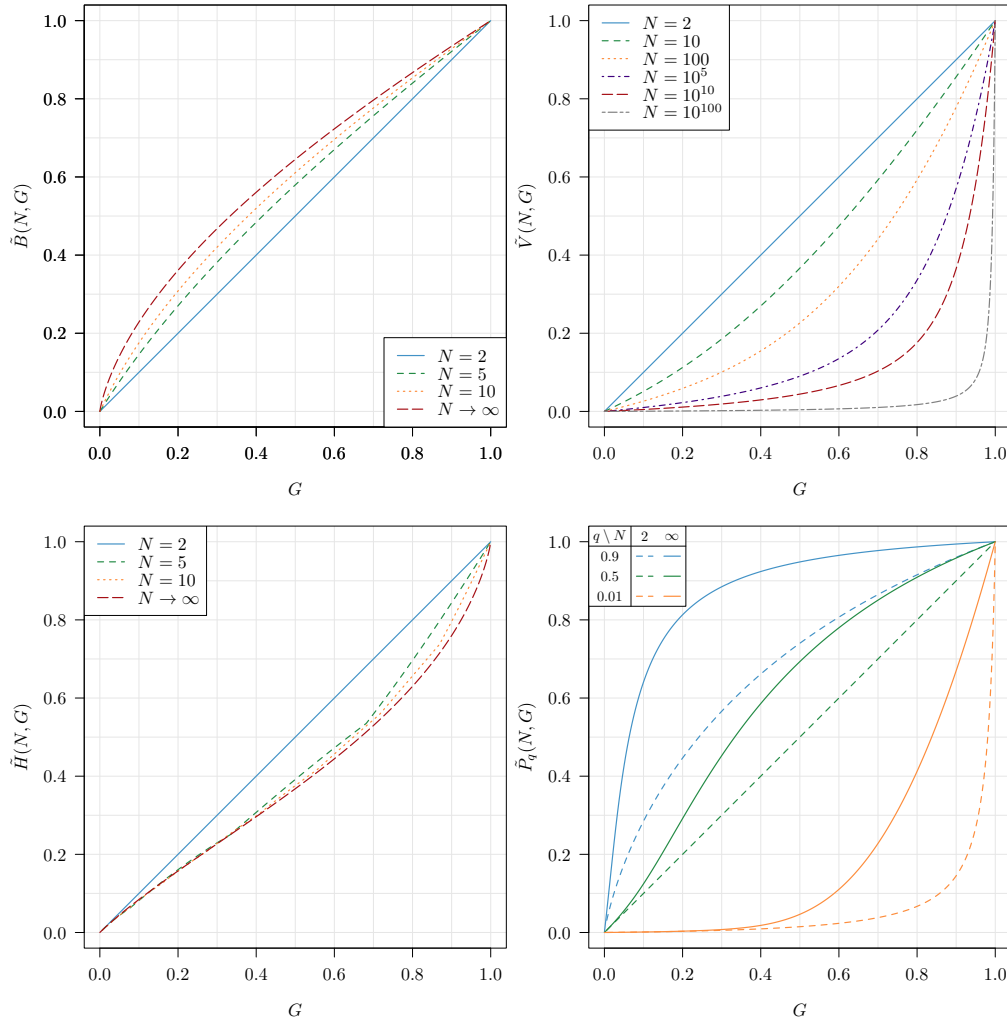


Figure 2: Functional dependence between the Gini index and other inequality measures for different sample sizes N in our model. The indexes are one-to-one functions of one another. Therefore, similar plots could be drawn for V as a function of B , H as a function of P_q , etc.

4. Experiments

Looking beyond our simple iterative process, we know that uncountably many normalised vectors yield a specific Gini, Bonferroni, or any other index. After all, these measures were introduced to respond to the different needs of the practitioners (Imedio-Olmedo et al., 2012; McVinish and Lester, 2020). Some of them are, for example, more responsive to the increasing of the amount of mass in the tail of the distribution (via the principle of progressive transfers) than others. In particular, as reported by Ciommi et al. (2022), the Gini, Bonferroni, and De Vergottini indices belong to the class of linear measures introduced by Mehran (1976). They note that *for the Bonferroni and De Vergottini indices, the effect of a transfer also depends on the position of individuals, making the Bonferroni index more sensitive to transfers that occur at the lower end of the income distribution and the De Vergottini index more sensitive to variations among the richest*.

Theoretical models are merely approximations of the real-world phenomena under scrutiny, but some models are more useful than others. Bertoli-Barsotti et al. (2024) have already noted that the Gini-stable model provides a good fit to a variety of informetric and other data, even though the assumption about the Gini index's being constant throughout all iterations is merely an idealisation, which does not necessarily have to hold throughout the whole agent's lifespan.

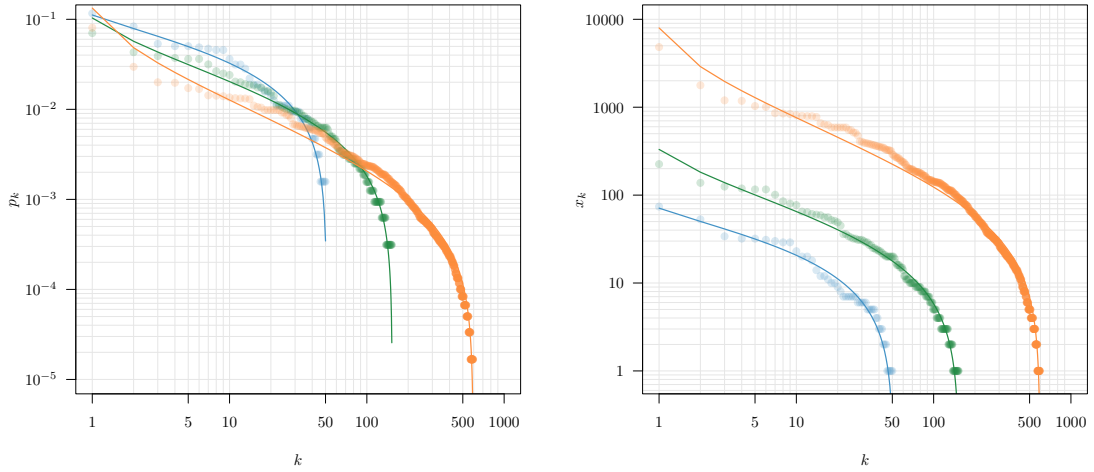


Figure 3: Three example citation vectors (left: normalised, right: original) and the corresponding fitted models (Eq. (1)); note the log-log scale. We note a very good fit in each case.

Therefore, we should be interested in verifying how well our formulae approximate the hidden relationships present in real datasets. Let us thus consider citation data from the RePEc database (Research Papers in Economics; see <https://citec.repec.org/>), which features 66,347 authors and 1,843,967 papers. In the data cleansing step, we have omitted the authors who published less than 5 cited papers and whose h -index was less than 3 for such samples are too small to make the analysis robust enough. This resulted in $n = 36,425$ citation records of the form $(x_1^{(i)}, \dots, x_{N^{(i)}}^{(i)})$, where $N^{(i)}$ gives the total number of items published by the i -th author and $x_k^{(i)}$ gives the number of citations to their k -th most cited works.

Figure 3 shows three example (quite representative of the whole database) vectors from the RePEc database: observed (points) and predicted (lines) values for p_k (left) and their non-normalised versions, x_k (right). We see a good fit over most parts of the data domain. Testing all vectors with the discrete Kolmogorov–Smirnov test (Arnold and Emerson, 2011) resulted in rejecting the null hypothesis that data follow our distribution only in 1% of the cases (at the significance level of 0.05). This comes as no surprise, as the proposed model is equivalent to the 3DI model (Siudem et al., 2020) and we have already seen its usefulness in the case of modelling citations to computer science papers. Nevertheless, we should note that the statistical power of the K-S test is not particularly high.

For each author, we computed their actual (observed) Gini index $\hat{G}^{(i)} = \mathcal{G}(p_1^{(i)}, \dots, p_{N^{(i)}}^{(i)})$ with $p_k^{(i)} = X_k^{(i)} / \sum_{j=1}^{N^{(i)}} X_j^{(i)}$. Knowing the value of $\hat{G}^{(i)}$, based on the derived formulae (see Eqs. (12)–(15)), we computed the predicted values of the indices: $\hat{B}(N, \hat{G}^{(i)})$, $\hat{V}(N, \hat{G}^{(i)})$, $\hat{V}_q(N, \hat{G}^{(i)})$ and $\hat{P}_q(N, \hat{G}^{(i)})$. This way, we can compare these approximated values with the observed ones, i.e., $\hat{B}^{(i)} = \mathcal{B}(p_1^{(i)}, \dots, p_{N^{(i)}}^{(i)})$, $\hat{V}^{(i)} = \mathcal{V}(p_1^{(i)}, \dots, p_{N^{(i)}}^{(i)})$, $\hat{H}^{(i)} = \mathcal{H}(p_1^{(i)}, \dots, p_{N^{(i)}}^{(i)})$, and $\hat{P}_q^{(i)} = \mathcal{P}(p_1^{(i)}, \dots, p_{N^{(i)}}^{(i)})$.

Recall that Figure 2 describes the theoretical relationships between the Gini index and other metrics. If, overall, our model describes the real data well, we should expect to see these dependencies in the case of the RePEc vectors too. Figure 4 presents a scatter plot of the values of different indices as functions of $\hat{G}$ for all vectors with $N > 200$ (the De Vergottini index was not included as it approaches 0 for large N s; see Remark 2). Ideally, they should lie close to the theoretical curves (depicted as well). And this is approximately the case.

Furthermore, Figure 5 presents similar results, but for vectors of lengths $N = 6$ (green), $N = 10$ (pink), and $N = 50 \pm 1$ (blue; the ± 1 part is to increase the number of data points). Additionally, we coloured the areas representing all of the possible values which could be obtained for vectors generated from our model. In other words, for any vector, we expect the pairs $(\hat{G}, \hat{B})$, $(\hat{G}, \hat{V})$, etc. to lie in the grey zone. This is true for the vast majority of the real data points.

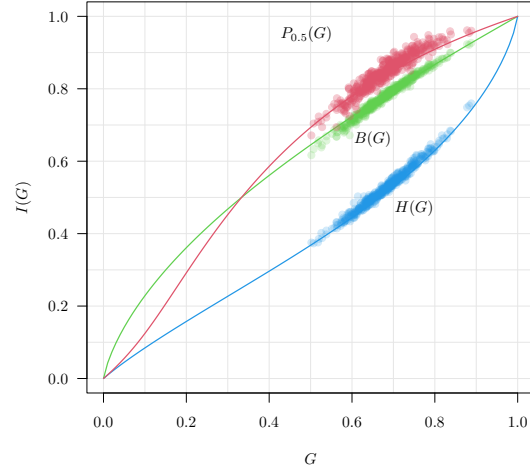


Figure 4: Inequality indices as functions of the Gini index for all RePEc citation vectors with $N > 200$ (points). They match the derived theoretical curves (lines) quite well, confirming the usefulness of our model.

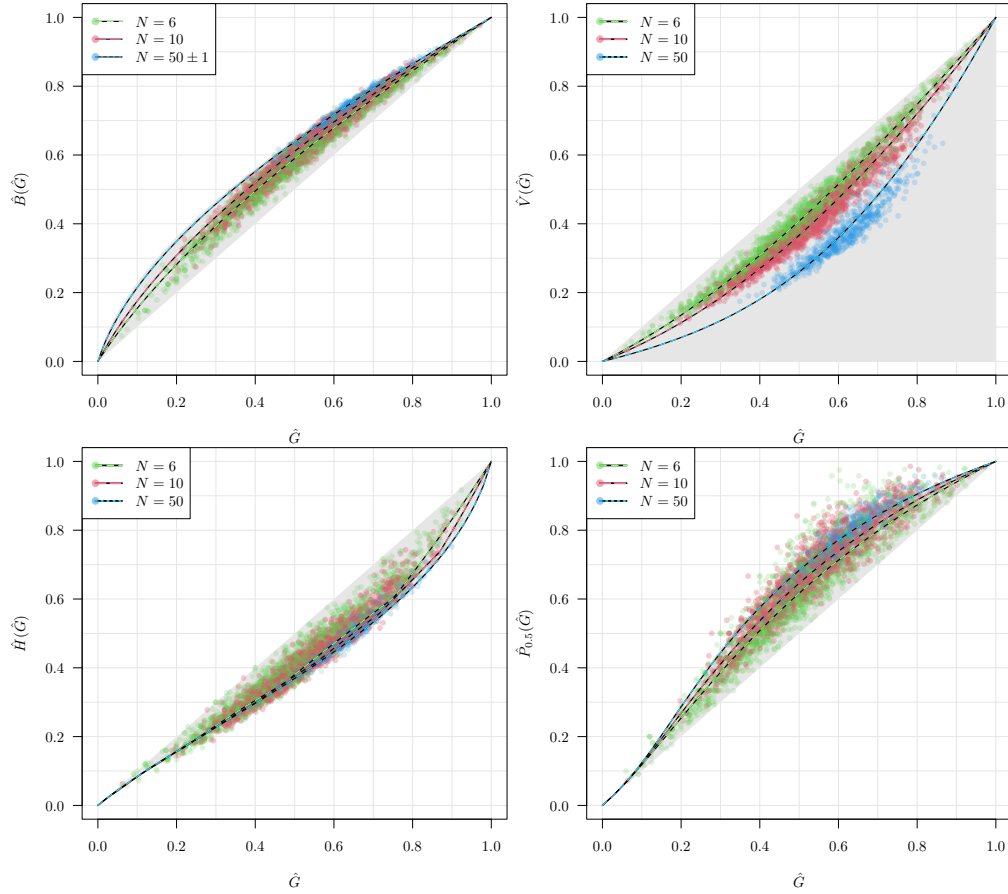


Figure 5: Predicted ($\tilde{B}(N, \hat{G})$ vs $\hat{G}$, $\tilde{V}(N, \hat{G})$ vs $\hat{G}$, etc.; thick curves) and observed ($\hat{B}$ vs $\hat{G}$ etc.; points) values of different inequality measures for the RePEc citation records of different lengths N . Grey areas represent the values which can be obtained from our model.

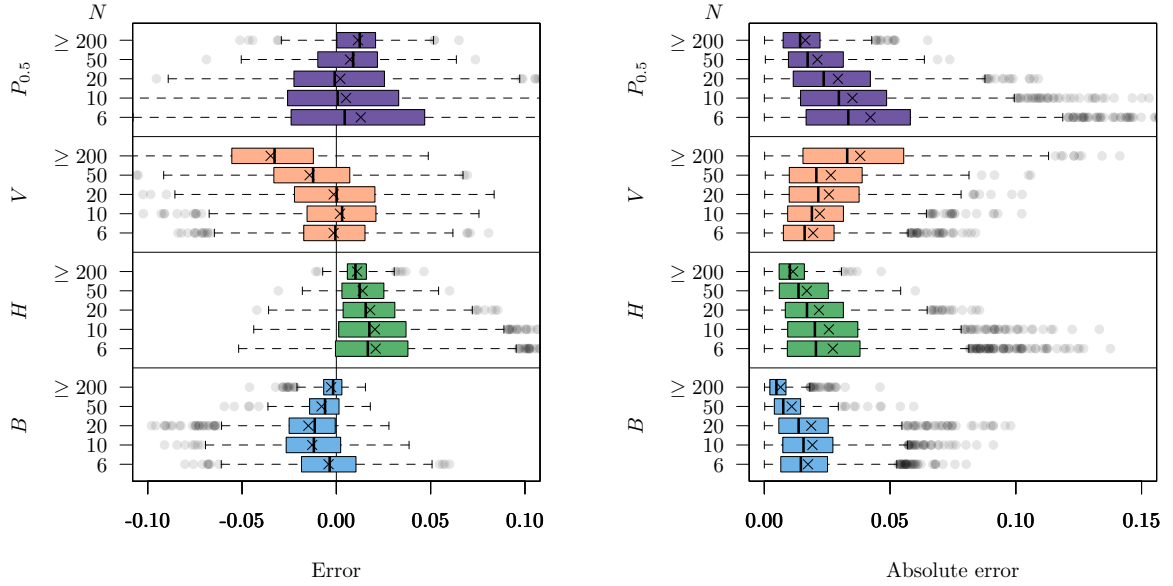


Figure 6: Box plots of prediction errors, i.e., $\hat{I} - \tilde{I}$ (left) and absolute prediction errors, i.e., $|\hat{I} - \tilde{I}|$ (right) for different values of N and different indices; RePEc data.

Table 1: Spearman correlation coefficients between index $\hat{I} \in \{\hat{H}, \hat{B}, \hat{V}, \hat{P}_{0.5}\}$ and index $\hat{G}$ calculated directly from the RePEc data for subsets of vectors of a given length and for all vectors (last row).

$N \setminus \hat{I}$	$\hat{H}$	$\hat{B}$	$\hat{V}$	$\hat{P}_{0.5}$
5	0.98	0.99	0.99	0.97
10	0.98	0.98	0.98	0.94
15	0.98	0.98	0.97	0.95
20	0.98	0.98	0.97	0.95
50	0.98	0.99	0.96	0.93
100	0.99	0.99	0.96	0.97
all	0.97	0.98	0.82	0.92

The prediction errors for different vector lengths are summarised in Figure 6. The absolute values of errors, i.e., $|\hat{I} - \tilde{I}|$ are usually less than 0.02–0.03. The errors themselves (bias) are small as well.

We can also be interested in the way they order the vectors of interest. To test if all indices give similar rankings, we can calculate the Spearman correlation coefficient between them. Table 1 presents the correlations between $\hat{G}$ and $\hat{I} \in \{\hat{H}, \hat{B}, \hat{V}, \hat{P}_{0.5}\}$ for subsets of vectors of a given length. For vectors of equal lengths, the obtained rankings are very similar. If we take a look at the whole data set (the last row), the correlations are still high and the biggest difference can be seen for index V . It is however not surprising as its dependence on G varies greatly for different values of N .

Overall, we obtained a good fit to our theoretical derivations. For data approximately following our model, we may approximately assume that when we consider vectors of similar lengths, the choice of the inequality measure is secondary, as the indices can be considered functions of one another. Therefore, in such a case, we can be faithful to the simplest indicator: the Gini index.

5. Conclusions

It may be interesting to note that we can easily generalise our Gini-stable model $i \rightarrow p_i^{(N,G)}$, $i = 1, 2, \dots, N$, to take the production factor into account by writing $\mathbf{y} = C\mathbf{p}^{(N,G)}$, $C > 0$. While $p_i^{(N,G)}$ may be interpreted as the relative number of items in the i -th cell of the IPP, $y_i = Cp_i^{(N,G)}$ represents its corresponding absolute value. For example, the vector $\mathbf{y}$ may express the number of citations earned by N articles. In that case, C is the total number of citations. Hence, both elements of the theory on impact by Egghe and Rousseau (2022, 2023a) are independently present in the array model $\mathbf{y} = C\mathbf{p}^{(N,G)}$.

The discussed process yields $\mathbf{p}^{(N,G)}$ that are totally ordered by the majorisation relation $\leq$. Following Shorrocks and Foster (1987) (but see also Marshall et al., 2011; Lambert, 2001; Chantreuil and Trannoy, 2011; Zheng, 2007; Bosmans, 2016), a function I is an index of inequality if and only if it is symmetric and strictly Schur convex.

Hence, for all ordered normalised N -vectors $\mathbf{p} \neq \mathbf{q}$, if $\mathbf{p} \leq \mathbf{q}$, then $I(\mathbf{p}) < I(\mathbf{q})$, where the direction of the inequality is uniquely determined by the Pigou–Dalton condition (e.g., Patty and Penn, 2019). In particular, for every vector $\mathbf{p}^{(N,G)}$, the parameter G can be interpreted as the (normalised) Gini index $\mathcal{G}(\mathbf{p}^{(N,G)}) = G$. This implies that $\mathcal{G}$ is an order preserving function (Marshall et al., 2011) on $\{\mathbf{p}^{(N,G)}\}$ with respect to G .

The main requirement to impose to a measure of concentration (inequality) to be “acceptable” is that to be coherent to the Lorenz ordering. By construction, the Gini index is acceptable because it is coherent with the Lorenz ordering (Egghe, 2010). But other indices are equally acceptable as measures of concentration. Examples of such indices are B , H , V and P_q . How can we compare them with each other? At least within the case of distributions belonging to the Gini-stable family, we showed that it is possible to derive their explicit expressions as one-to-one functions of G and N . This allows us to distinguish and compare all these measures of concentration according to their degree of sensitivity to inequality.

For data vectors of similar sizes that follow closely our model, the indices can be considered equivalent. An analysis of an empirical dataset (citation vectors in economics) confirmed our results. Of course, we need to keep in mind that our model is not universal: Bertoli-Barsotti et al. (2024) presented the instances of both fair and poor fits.

The paper focused on an analysis of the Gini-stable process and relations between inequality measures in it. Even though the empirical data fit the vectors generated by this model reasonably well, there are some questions that have not been answered yet. In particular, the analysed process assumes that the vectors have the same Gini index over their whole lifespan. The extent to which this idealisation is correct, at least approximately, is yet to be determined, although data that could be used to validate this hypothesis is difficult to obtain. We also assume that wealth is being distributed equally over time, which is also an assumption that should be tested on real data.

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Conflict of interest

The authors certify that they have no affiliations with or involvement in any organisation or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

CRedit authorship contribution statement

Lucio Bertoli-Barsotti: Conceptualisation of this study, Methodology, Writing. **Marek Gagolewski:** Conceptualisation of this study, Methodology, Data Curation, Investigation, Software, Writing. **Grzegorz Siudem:** Conceptualisation of this study, Methodology, Writing. **Barbara Żogała-Siudem:** Data Curation, Investigation, Visualisation, Software, Writing.

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References

- Arnold, B.C., 2015. Pareto Distributions. Chapman and Hall/CRC, New York, NY, USA. doi:10.1201/b18141.
- Arnold, T.A., Emerson, J.W., 2011. Nonparametric goodness-of-fit tests for discrete null distributions. *The R Journal* 3, 34–39. doi:10.32614/RJ-2011-016.
- Beliakov, G., Gagolewski, M., James, S., 2016. Penalty-based and other representations of economic inequality. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* 24(Suppl.1), 1–23. doi:10.1142/S0218488516400018.
- Bertoli-Barsotti, L., 2023. Equivalent Gini coefficient, not shape parameter! *Scientometrics* 128, 867–870. doi:10.1007/s11192-022-04571-8.
- Bertoli-Barsotti, L., Gagolewski, M., Siudem, G., Żogała Siudem, B., 2024. Gini-stable Lorenz curves and their relation to the generalised Pareto distribution. *Journal of Informetrics* 18, 101499. doi:10.1016/j.joi.2024.101499.
- Bonferroni, C., 1930. *Elementi di statistica generale*. Libreria Seber, Firenze.
- Bosmans, K., 2016. Consistent comparisons of attainment and shortfall inequality: A critical examination. *Health Economics* 25, 1425–1432.
- Brin, S., Page, L., 1998. The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems* 30, 107–117. doi:10.1016/S0169-7552(98)00110-X.
- Cena, A., Gagolewski, M., Siudem, G., Żogała Siudem, B., 2022. Validating citation models by proxy indices. *Journal of Informetrics* 16, 101267. doi:10.1016/j.joi.2022.101267.
- Chantreuil, F., Trannoy, A., 2011. Inequality decomposition values. *Annals of Economics and Statistics/Annales d'Économie et de Statistique* 101/102, 13–36.
- Ciommi, M., Gigliarano, C., Giorgi, G., 2022. Bonferroni and de Vergottini are back: New subgroup decompositions and bipolarization measures. *Fuzzy Sets and Systems* 433, 22–53.
- Cowell, F.A., 2000. Measurement of inequality, in: Atkinson, A., Bourguignon, F. (Eds.), *Handbook of Income Distribution*. Elsevier. volume 1, pp. 87–166.
- Egghe, L., 2005. Power laws in the information production process: Lotkaian informetrics. Emerald Group Publishing Limited.
- Egghe, L., 2010. The Hirsch index and related impact measures. *Annu. Rev. Inf. Sci. Technol.* 44, 65–114.
- Egghe, L., Rousseau, R., 1990. *Introduction to informetrics. Quantitative methods in library, documentation and information science*. Elsevier Science Publishers.
- Egghe, L., Rousseau, R., 1991. Transfer principles and a classification of concentration measures. *Journal of the American Society for Information Science* 42, 479–489.
- Egghe, L., Rousseau, R., 2001. Symmetric and asymmetric theory of relative concentration and applications. *Scientometrics* 52, 261–290.
- Egghe, L., Rousseau, R., 2022. Rank-frequency data and impact in a continuous model: Introducing impact bundles. *Journal of Informetrics* 16, 101297.
- Egghe, L., Rousseau, R., 2023a. Global impact measures. *Scientometrics* 128, 699–707.
- Egghe, L., Rousseau, R., 2023b. Global informetric impact: A description and definition using bundles. *Journal of Informetrics* 17, 101366.
- Gagolewski, M., Żogała Siudem, B., Siudem, G., Cena, A., 2022. Ockham's index of citation impact. *Scientometrics* 127, 2829–2845. doi:10.1007/s11192-022-04345-2.
- Gini, C., 1912. Variabilità e mutabilità. C. Cuppini, Bologna.
- González-Pereira, B., Guerrero-Bote, V.P., Moya-Anegón, F., 2010. A new approach to the metric of journals' scientific prestige: The sjr indicator. *Journal of informetrics* 4, 379–391.
- Holme, P., 2022. Universality out of order. *Nature Communications* 13, 2355.
- Hoover, E., 1941. Interstate redistribution of population, 1850–1940. *The Journal of Economic History* 1, 199–205.
- Imedio-Olmedo, L., Parrado-Gallardo, E., Bárcena-Martín, E., 2012. Income inequality indices interpreted as measures of relative deprivation/satisfaction. *Social Indicators Research* 109, 471–491.
- Iñiguez, G., Pineda, C., Gershenson, C., Barabási, A.L., 2022. Dynamics of ranking. *Nature communications* 13, 1646.
- Lambert, P.J., 2001. *The distribution and redistribution of income* (3rd Edition). Manchester University Press.
- Marshall, A.W., Olkin, I., 2007. *Life distributions: Structure of Nonparametric, Semiparametric, and Parametric Families*. Springer.
- Marshall, A.W., Olkin, I., Arnold, B.C., 2011. *Inequalities: Theory of majorization and its applications* (2nd Edition). Springer Science Business Media.
- McVinish, R., Lester, R., 2020. Measuring aggregation in parasite populations. *Journal of the Royal Society Interface* 17, 20190886. doi:10.1098/rsif.2019.0886.
- Mehran, F., 1976. Linear measures of income inequality. *Econometrica* 44, 805–809.
- Newman, M., 2005. Power laws, Pareto distributions and Zipf's law. *Contemporary Physics* 46, 323–351. doi:10.1080/00107510500052444.
- Patty, J.W., Penn, E.M., 2019. Measuring fairness, inequality, and big data: Social choice since Arrow. *Annual Review of Political Science* 22, 435–460. doi:10.1146/annurev-polisci-022018-024704.
- Perc, M., 2014. The Matthew effect in empirical data. *Journal of The Royal Society Interface* 11. doi:10.1098/rsif.2014.0378.
- Petersen, A.M., Stanley, H.E., Succi, S., 2011. Statistical regularities in the rank-citation profile of scientists. *Scientific Reports* 1, 181. doi:10.1038/srep00181.
- Pickands III, J., 1975. Statistical inference using extreme order statistics. *The Annals of Statistics* , 119–131.
- Prathap, G., 2014. The zynergy-index and the formula for the h-index. *Journal of the Association for Information Science and Technology* 65, 426–427.
- Prathap, G., 2022. Comments on the paper of Gagolewski et al.: Ockham's index of citation impact. *Scientometrics* 127, 6051–6054.
- Price, D., 1965. Networks of scientific papers. *Science* 149, 510–515. doi:10.1126/science.149.3683.510.
- Rousseau, R., 1992. Concentration and diversity of availability and use in information systems: A positive reinforcement model. *Journal of the American Society for Information Science* 43, 391–395.
- Rousseau, R., 2011. Lorenz curves determine partial orders for comparing network structures. *DESIDOC Journal of Library & Information*

Technology 31.

- Rousseau, R., Egghe, L., Guns, R., 2018. *Becoming metric-wise: A bibliometric guide for researchers*. Chandos Publishing.
- Shorrocks, A.F., Foster, J.E., 1987. Transfer sensitive inequality measures. *The Review of Economic Studies* 54, 485–497.
- Silber, J., 2012. *Handbook of income inequality measurement*. volume 71. Springer Science & Business Media.
- Singh, C.K., Barme, E., Ward, R., Tupikina, L., Santolini, M., 2022. Quantifying the rise and fall of scientific fields. *PLoS ONE* 17, e0270131.
- Siudem, G., Nowak, P., Gagolewski, M., 2022. Power laws, the Price model, and the Pareto type-2 distribution. *Physica A: Statistical Mechanics and its Applications* 606, 128059. doi:10.1016/j.physa.2022.128059.
- Siudem, G., Żogała-Siudem, B., Cena, A., Gagolewski, M., 2020. Three dimensions of scientific impact. *Proceedings of the National Academy of Sciences* 117, 13896–13900. doi:10.1073/pnas.2001064117.
- Vergottini, M.D., 1940. Sul significato di alcuni indici di concentrazione. *Giornale degli Economisti e Annali di Economia* 11, 317–347.
- Vergottini, M.D., 1950. Sugli indici di concentrazione. *Statistica* 10, 445–454.
- Voitalov, I., van der Hoorn, P., van der Hofstad, R., Krioukov, D., 2019. Scale-free networks well done. *Physical Review Research* 1, 033034.
- Yitzhaki, S., Schechtman, E., 2013. *The Gini methodology: A primer on a statistical methodology*. Springer.
- Zheng, B., 2007. Unit-consistent decomposable inequality measures. *Economica* 74, 97–111.